

MonteCarlo: From Classic to Quantum

I. Classical MarkovChain Monte Carlo Algorithms

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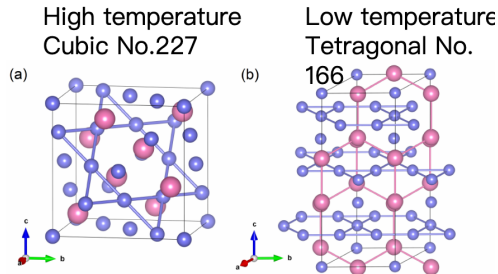
Outline

- 1 Motivation: TbCo_2 and the Kondo-Ising Model
- 2 Monte Carlo Fundamentals
- 3 Classical Algorithms: Metropolis, SW, and Wolff
- 4 Wolff Algorithm on Infinite Lattice
- 5 Loop and Worm Algorithms for Quantum Systems
- 6 Summary and Outlook

TbCo₂: A Ferrimagnetic Compound

- Co atoms form a network of corner-sharing tetrahedra
- Tb occupies the larger interstitial sites
- Ferrimagnetic order: Tb moments ($\sim 9 \mu_B$) and Co moments ($\sim 1 \mu_B$) couple *antiparallel*

Ref: Applied Physics Letters **121**, 232404 (2022)

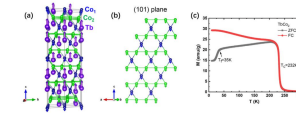


Non-collinear ferrimagnetic structure
under intermediate field $0 < \mu_0 H < \mu_0 H_s$

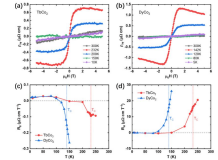
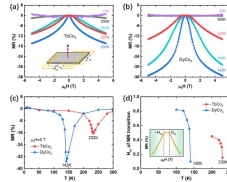
Transport Properties of TbCo₂

- Non-linear Hall resistivity ρ_{xy} — characteristic of **Anomalous Hall Effect (AHE)**
- **Sign Flip of Anomalous Hall and carrier** observed near $T_C = 232$ K
- We want to explain the high T_C of magnetism and the origin of these transport signatures

□ Polycrystalline samples results



- Plateau-like magnetoresistance



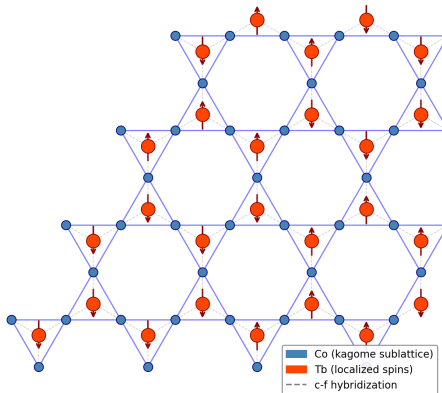
Appl. Phys. Lett. 121, 232404 (2022)

Transport measurements on polycrystalline TbCo₂
(Appl. Phys. Lett. **121**, 232404)

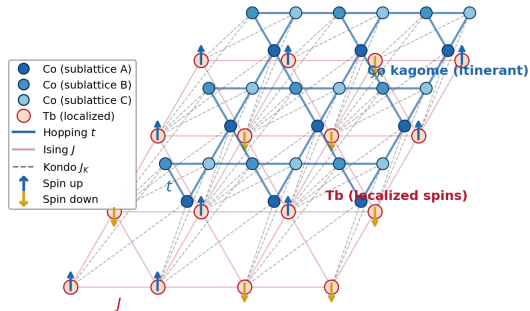
From TbCo₂ to the Kondo-Ising Model

Kondo-Ising Model on Kagome Lattice

$$H = -t \sum_{\langle ij \rangle \sigma} c_{i\sigma}^\dagger c_{j\sigma} + J_K \sum_i S_i^z S_i^z - J \sum_{\langle ij \rangle} S_i^z S_j^z$$



2D kagome lattice view



$$H = -t \sum_{\langle ij \rangle \sigma} c_{i\sigma}^\dagger c_{j\sigma} + J_K \sum_i S_i^z S_i^z - J \sum_{\langle ij \rangle} S_i^z S_j^z$$

Isometric bilayer view

Kondo-Ising Model: Hamiltonian

Kondo-Ising Hamiltonian

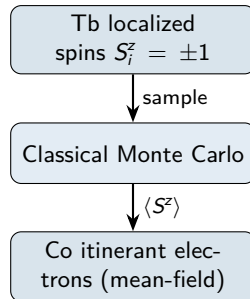
$$H = \underbrace{-t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma}}_{\text{itinerant electrons (Co)}} + \underbrace{J_K \sum_i S_i^z s_i^z}_{\text{Kondo coupling}} - \underbrace{J \sum_{\langle ij \rangle} S_i^z S_j^z}_{\text{Ising interaction (Tb)}}$$

- **Assumption:** c-f hybridization is weak $\rightarrow$ project onto Ising anisotropy limit
- The model separates naturally into two parts:
 - ① **Itinerant electron part (Co):** temperature enters only through Fermi-Dirac smearing
 - ② **Localized spin part (Tb):** treated as a classical Ising system via **Monte Carlo**
- The localized spins act as an effective **mean field** on itinerant electrons

Refs: W.-W. Yang *et al.*, PRB **100**, 045148 (2019); PRB **102**, 195141 (2020); PRB **104**, 165146 (2021)

Roadmap: Why Classical Monte Carlo?

- The Kondo-Ising model requires **thermal sampling** of the localized spin configuration S_i^z at each temperature
- This is a **classical** Monte Carlo problem:
 - Configuration space: $\{S_i^z = \pm 1\}$
 - Boltzmann weight: $e^{-\beta E[\{S^z\}]}$
- **This lecture:** classical MC algorithms (Metropolis, SW, Wolff)
- **Later lectures:** apply to Kondo-Ising; extend to quantum MC



Monte Carlo Sampling: The Core Idea

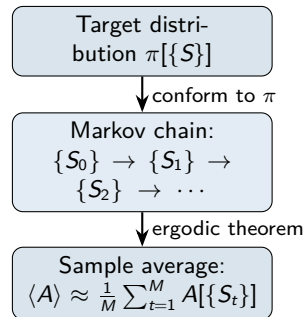
- **Goal:** compute thermal expectation values

$$\langle A \rangle = \sum_{\{S\}} A[\{S\}] \pi[\{S\}]$$

- Boltzmann distribution:

$$\pi[\{S\}] = \frac{e^{-\beta \mathcal{H}[\{S\}]}}{Z}, \quad Z = \sum_{\{S\}} e^{-\beta \mathcal{H}[\{S\}]}$$

- **Problem:** Z is intractable — combinatorial explosion of configurations
- **Solution:** construct a Markov chain that visits configurations with frequency $\propto \pi$



Two Conditions for Correct Sampling

A Markov chain sampler must satisfy two conditions to converge to the target distribution π :

1. Detailed Balance

$$W(A \rightarrow B) \pi(A) = W(B \rightarrow A) \pi(B)$$

Equivalently: $\frac{W(A \rightarrow B)}{W(B \rightarrow A)} = e^{-\beta(E_B - E_A)}$
 $\Rightarrow \pi$ is a **stationary distribution** of the chain.

2. Ergodicity

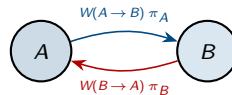
Every configuration is reachable from any other in finitely many steps:

$$\forall A, B: \exists n \geq 1, \text{Prob}(A \xrightarrow{n} B) > 0$$

$\Rightarrow$ the stationary distribution is **unique**.

Why They Suffice

- 1 Detailed balance $\Rightarrow \pi$ is a **fixed point**
- 2 Ergodicity $\Rightarrow$ fixed point is **unique**
- 3 Together $\Rightarrow$ chain converges to π from *any* starting state



Net flow = 0 at equilibrium

Thermalization: When Can We Start Measuring?

Thermalization (Burn-in)

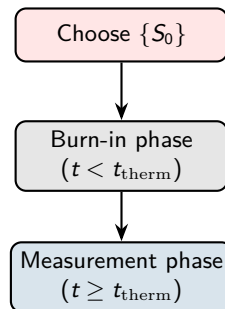
The Markov chain needs t_{therm} steps before samples follow π . Discard the first t_{therm} samples before computing observables.

Practical checks:

- 1 Start from *different* initial configurations (e.g. all spins up vs. random — see previous slide)
- 2 Track $\langle m \rangle$, $\langle E \rangle$, ... until they converge
- 3 Confirm that **independent** runs agree

Advanced: the *algorithmic overlap* q between two replicas provides a thermodynamic-level indicator of equilibration.

Ref: arXiv:2604.10254; detailed discussion in Sec. 7.

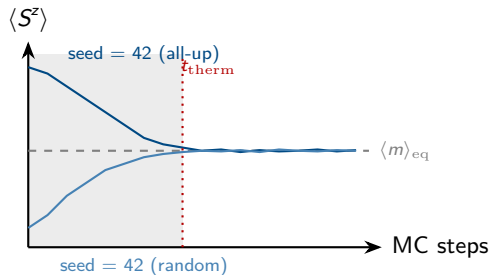


Discard burn-in data;
measure from here

A view of thermalization from two different initial states

- Choose two different initial configurations (e.g. all spins up vs. random)

Whether an MC algorithm is stable and how to characterize its trajectory will be discussed in Lecture 2.



The 2D Ising Model: Testbed for Classical Algorithms

Hamiltonian

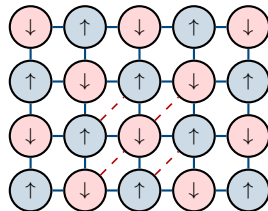
$$\mathcal{H} = -J \sum_{\langle ij \rangle} S_i^z S_j^z$$

- $S_i^z = \pm 1$ on a 2D square lattice
- $J > 0$: ferromagnetic; $J < 0$: antiferromagnetic

With next-nearest-neighbor frustration

$$\mathcal{H} = -J \sum_{\langle ij \rangle} S_i^z S_j^z + J' \sum_{\langle\langle ik \rangle\rangle} S_i^z S_k^z$$

$J' > 0$: competing AFM interaction $\Rightarrow$ frustration



NN: J (solid)

NNN: J' (dashed)

Metropolis Algorithm: Single-Spin-Flip

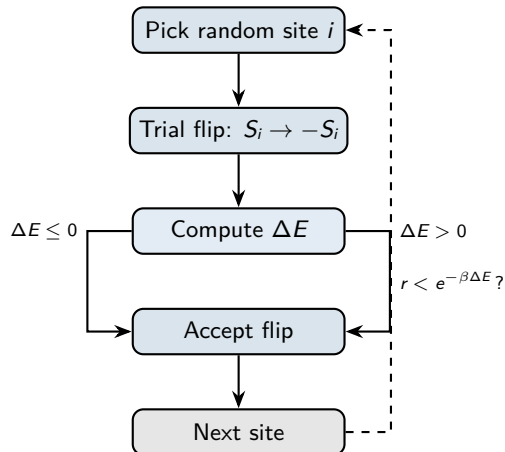
Acceptance Probability

$$W(S \rightarrow S') = \min(1, e^{-\beta \Delta E})$$

where $\Delta E = E(S') - E(S)$ is the energy change from flipping one spin.

Properties:

- **Local** update: flip one spin at a time
- Downhill moves ($\Delta E < 0$): always accepted
- Uphill moves: accepted with probability $e^{-\beta \Delta E}$
- Ergodic: any configuration reachable by sequential flips



Metropolis: Detailed Balance Verification

Proof of Detailed Balance

For a single spin flip at site i , let S' differ from S only at i :

$$\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = \frac{\min(1, e^{-\beta(E_{S'} - E_S)})}{\min(1, e^{-\beta(E_S - E_{S'})})}$$

Case 1: $E_{S'} \leq E_S \Rightarrow$ numerator = 1, denominator = $e^{-\beta(E_S - E_{S'})}$

$$\therefore \frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = e^{-\beta(E_{S'} - E_S)} = \frac{\pi(S')}{\pi(S)} \quad \checkmark$$

Case 2: $E_{S'} > E_S \Rightarrow$ same result by symmetry.

Ergodicity

Since each spin can be flipped individually, any configuration $\{S\}$ can be transformed into any other $\{S'\}$ by at most N single-spin flips. $\checkmark$

Swendsen–Wang Algorithm: Bond Percolation

Fortuin–Kasteleyn Bond Activation

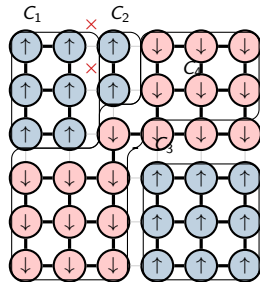
For each pair of **aligned** nearest neighbors $S_i = S_j$:

$$p_{\text{bond}} = 1 - e^{-2\beta J}$$

Activate bond with probability p_{bond} ; never bond anti-aligned pairs.

Algorithm:

- 1 Scan all NN bonds; activate with prob p_{bond}
- 2 Identify connected clusters via union-find
- 3 Flip each cluster independently with prob $1/2$



Swendsen-Wang Flipping Process

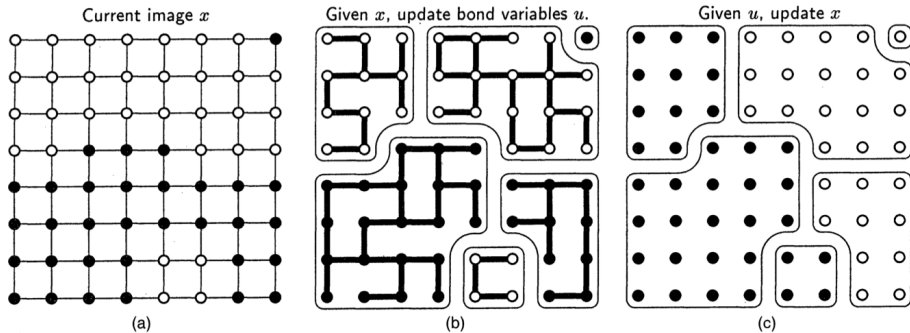


Figure 2. The Swendsen–Wang Algorithm for the Ising Model on the 8×8 Lattice. (a) Initial image x and Markov random field graph for $\pi(x)$. (b) Given the current image x , the bond variables u are generated uniformly over the interval $(0, e^{\beta I[x_i=x_j]})$. If $u_{ij} > 1$ (marked by the thick lines), x_i is constrained to equal x_j . These constraints partition the image into clusters of like-colored sites. Clusters induced by this realization of $u|x$ are outlined. The Markov random field graph for $x|u$ differs from that of x , marginally; the auxiliary vector u strengthens the dependence between some neighboring sites, while completely removing it from others. (c) Given the bond variables u , x is now a coarse image of independent clusters. Because there is no external field in this example, each cluster is recolored black or white with probability .5.

Swendsen–Wang: Detailed Balance

Transition Probability

Probability to go from S to a specific S' :

$$W_{\text{SW}}(S \rightarrow S') = \sum_{B \sim S, S'} P(B | S) \left(\frac{1}{2}\right)^{N_C(B)}$$

Sum over bond configs B compatible with both S and S' ; each of N_C clusters flipped independently with prob $1/2$.

Detailed Balance Holds

In the FK representation ($e^{\beta J|E|}$ cancels in the ratio):

$$\frac{W_{\text{SW}}(S \rightarrow S')}{W_{\text{SW}}(S' \rightarrow S)} = \frac{\prod_{\langle ij \rangle} e^{\beta J(1+S'_i S'_j)}}{\prod_{\langle ij \rangle} e^{\beta J(1+S_i S_j)}} = e^{-\beta(E_{S'} - E_S)} = \frac{\pi(S')}{\pi(S)} \quad \checkmark$$

Key: $p = 1 - e^{-2\beta J}$ is chosen so that $e^{\beta J}(1 - p) = e^{-\beta J}$.

Ergodicity: with prob $(1-p)^{|E|} > 0$, all bonds absent $\Rightarrow N$ single-site clusters $\Rightarrow$ any config reachable. $\checkmark$

SW Detailed Balance: Full Derivation (*backup*)

Step 1. Bond probability:

$$P(B | S) = \prod_{\langle ij \rangle: S_i=S_j} p^{b_{ij}} (1-p)^{1-b_{ij}}$$

$b_{ij} \in \{0, 1\}$: bond indicator in B ; bonds only between aligned spins.

Step 2. Boltzmann weight (FK form):

$$\pi(S) \propto \prod_{\langle ij \rangle} e^{\beta J(1+S_i S_j)}$$

$\Rightarrow e^{2\beta J}$ for aligned, 1 for anti-aligned.

Step 3. Transition probability:

$$W_{\text{SW}}(S \rightarrow S') = \sum_{B \sim S, S'} P(B | S) \left(\frac{1}{2}\right)^{N_c}$$

Step 4. Detailed balance via key lemma
(next slide: $\pi(S) P(B|S) = \pi(S') P(B|S')$):

$$\begin{aligned} \pi(S) W(S \rightarrow S') &= \sum_B \underbrace{\pi(S) P(B|S)}_{= \pi(S') P(B|S')} (1/2)^{N_c} \\ &= \pi(S') W(S' \rightarrow S) \end{aligned}$$

$$\Rightarrow W(S \rightarrow S') / W(S' \rightarrow S) = \pi(S') / \pi(S) \checkmark$$

SW Detailed Balance: Why It Really Works (*backup*)

Key lemma. Define $f(S, B) = \pi(S) \cdot P(B \mid S)$:

$$f(S, B) \propto \prod_{\langle ij \rangle} e^{\beta J S_i S_j} \cdot p^{b_{ij}} (1-p)^{1-b_{ij}}$$

Per-bond contribution to f :

$$\begin{cases} e^{\beta J} \cdot p & b_{ij} = 1, S_i = S_j \\ e^{\beta J} (1-p) \stackrel{!}{=} e^{-\beta J} & b_{ij} = 0, S_i = S_j \\ e^{-\beta J} & b_{ij} = 0, S_i \neq S_j \end{cases}$$

$$e^{\beta J} (1-p) = e^{-\beta J} \text{ iff } p = 1 - e^{-2\beta J} \checkmark$$

Cluster flip invariance.

When cluster C_k flips ($S_i \rightarrow -S_i$ for $i \in C_k$):

- **Intra-cluster:** both spins flip $\Rightarrow S_i S_j$ unchanged $\Rightarrow$ contribution unchanged
- **Boundary** ($b_{ij}=0$, one site in C_k): $S_i S_j$ flips sign, but aligned $\leftrightarrow$ anti-aligned both give $e^{-\beta J} \Rightarrow$ unchanged

$\therefore f(S, B) = f(S', B)$ for every compatible B .

Detailed balance:

$$\begin{aligned} \pi(S) W(S \rightarrow S') &= \sum_B f(S, B) \left(\frac{1}{2}\right)^{N_c} \\ &= \sum_B f(S', B) \left(\frac{1}{2}\right)^{N_c} = \pi(S') W(S' \rightarrow S) \end{aligned}$$

Wolff Algorithm: Single-Cluster Update

Core Idea

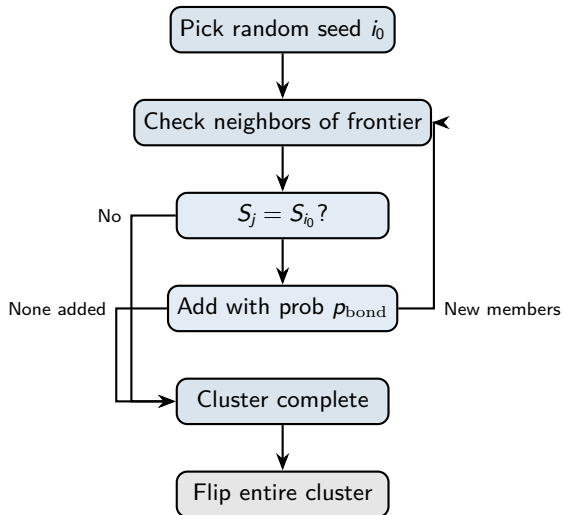
Instead of building *all* clusters (SW), grow **one cluster** from a random seed site using the same FK bond probability:

$$p_{\text{bond}} = 1 - e^{-2\beta J}$$

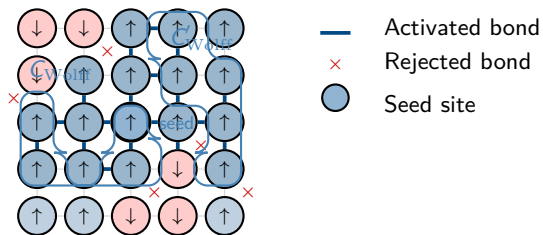
Algorithm:

- 1 Pick a random seed site i_0
- 2 For each unvisited neighbor j of the cluster frontier:
if $S_j = S_{i_0}$, add j with prob p_{bond}
- 3 Repeat step 2 until no new members
- 4 Flip the entire cluster

Ref: U. Wolff, PRL **62**, 361 (1989)



Wolff: Cluster Growth Example



Wolff: Detailed Balance

Transition Probability

$W(S \rightarrow S') \neq 0$ only when S' is obtained by flipping **one** connected cluster C of aligned spins. For such a pair:

$$W_{\text{Wolff}}(S \rightarrow S') = \frac{|C|}{N} \prod_{\mathcal{I}(C)} p \prod_{\substack{\partial C \\ S_i = S_j}} (1 - p)$$

$|C|/N$: seed in C ; $\mathcal{I}(C)$: internal bonds activated; ∂C : boundary bonds rejected (stop growth).

Detailed Balance

$|C|/N$ and $\prod_{\mathcal{I}} p$ cancel in ratio (same cluster, same internal bonds). Only ∂C contributes:

$$\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = \prod_{\substack{\langle ij \rangle \in \partial C \\ S_i = S_j}} \frac{(1 - p)}{1} \cdot \prod_{\substack{\langle ij \rangle \in \partial C \\ S_i \neq S_j}} \frac{1}{(1 - p)}$$

After flip: aligned $\leftrightarrow$ anti-aligned swap on ∂C . With $(1 - p) = e^{-2\beta J}$: $\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = e^{-\beta(E_{S'} - E_S)} = \frac{\pi(S')}{\pi(S)} \checkmark$

Ergodicity: cluster of size 1 (all bonds reject) $\Rightarrow$ single spin flip. $\checkmark$

Wolff Detailed Balance: Full Derivation (*backup*)

Key difference from SW: one Wolff step flips only *one* cluster C . $W(S \rightarrow S') = 0$ unless S' differs from S by flipping a single connected cluster. Arbitrary $S \rightarrow S'$ may need *multiple* Wolff steps — but detailed balance is checked per step.

Step 1. For S' reachable via cluster C :

$$W(S \rightarrow S') = \frac{|C|}{N} \prod_{\mathcal{I}(C)} p \prod_{\substack{\langle ij \rangle \in \partial C \\ S_i = S_j}} (1-p)$$

$\frac{|C|}{N}$: seed in C ; $\mathcal{I}(C)$: activated; ∂C : rejected.

Step 2. Cancel common factors:

$$\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = \frac{\prod_{\partial, S_i = S_j} (1-p)}{\prod_{\partial, S'_i = S'_j} (1-p)}$$

$|C|/N$ cancels (same C); $\prod_{\mathcal{I}} p$ cancels.

Step 3. Boundary bond contributions:

Each boundary bond ($i \in C, j \notin C$) contributes differently:

Bond in S	$W(S \rightarrow S')$	In S'	$W(S' \rightarrow S)$
Aligned	$(1-p)$ (reject)	Anti-aligned	1 (no check)
Anti-aligned	1 (no check)	Aligned	$(1-p)$ (reject)

After flip: aligned $\leftrightarrow$ anti-aligned swap $\Rightarrow$ contributions also swap.

Wolff: Verify Detailed Balance (*backup, cont.*)

Step 4. Substitute $1 - p = e^{-2\beta J}$:

$$\frac{W(S \rightarrow S')}{W(S' \rightarrow S)} = \prod_{\substack{\partial C \\ S_i = S_j}} e^{-2\beta J} \cdot \prod_{\substack{\partial C \\ S_i \neq S_j}} e^{2\beta J}$$

Let $n_{\uparrow}$ = aligned boundary bonds, $n_{\downarrow}$ = anti-aligned:

$$= (e^{-2\beta J})^{n_{\uparrow}} (e^{2\beta J})^{n_{\downarrow}} = e^{-2\beta J(n_{\uparrow} - n_{\downarrow})}$$

Step 5. Energy change on ∂C :

$$E_S|_{\partial C} = -J n_{\uparrow} + J n_{\downarrow}$$

$$E_{S'}|_{\partial C} = -J n_{\downarrow} + J n_{\uparrow}$$

(aligned $\leftrightarrow$ anti-aligned swap after flip)

$$\Delta E = E_{S'} - E_S = 2J(n_{\uparrow} - n_{\downarrow})$$

Conclusion

$$e^{-\beta \Delta E} = e^{-2\beta J(n_{\uparrow} - n_{\downarrow})} = \frac{W(S \rightarrow S')}{W(S' \rightarrow S)} \quad \checkmark$$

Comparison: Metropolis vs. SW vs. Wolff

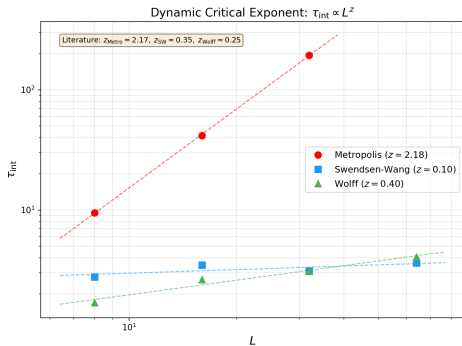
	Metropolis	Swendsen–Wang	Wolff
Update scope	Single spin	All clusters	One cluster
Clusters/sweep	N (trivial)	$\sim N$	1
Dynamic exp. z	≈ 2	≈ 0.35	≈ 0.25
Cost/flip	$O(1)$	$O(N)$	$O(\langle C \rangle)$
Best for	Generic models	Frustrated systems	Ferromagnets

- **Metropolis**: simplest; universal but suffers *critical slowing down* ($z \approx 2$)
- **SW**: global update via FK percolation; eliminates most critical slowing down
- **Wolff**: single-cluster variant; even lower z than SW for ferromagnets
- All three satisfy detailed balance & ergodicity
- Near T_c : Wolff is typically the most *efficient per unit cost*

Dynamic Critical Exponent: Numerical Measurement

Definition

The **integrated autocorrelation time** τ_{int} of an observable (here $|m|$) scales with system size near T_c as $\tau_{\text{int}} \propto L^z$, where z is the **dynamic critical exponent**. Measured from time series at T_c for $L = 8, 16, 32, 64$.



Cluster algorithms ($z \sim 0.2$ – 0.4) eliminate most of the critical slowing down that plagues Metropolis ($z \sim 2$).

(Section 5 — to be written)

(Section 6 — to be written)

(Section 7 — to be written)